

直流无刷无齿槽电机规格书  
SLOTLESS BRUSHLESS DC MOTOR SPECIFICATIONS

受 控 文 件

上海鸣志电器股份有限公司

ECU16052H24-S002+PG16M-0172

| 规格 Specifications        |                                         | Values       | Tolerance | Unit    |
|--------------------------|-----------------------------------------|--------------|-----------|---------|
| 1. 电机参数 Motor parameters |                                         |              |           |         |
| 1.1                      | 轴承类型 Bearings type of motor             | Ball bearing |           |         |
| 1.2                      | 电机极数 Number of poles                    | 2            |           |         |
| 1.3                      | 相数 Number of phases                     | 3            |           |         |
| 1.4                      | 绝缘等级 Insulation class                   | F (155°C)    |           |         |
| 1.5                      | 相间电阻 Terminal resistance phase to phase | 3.16         | ±8%       | Ω       |
| 1.6                      | 相间电感 Terminal inductance phase to phase | 253          | Typical   | μH      |
| 1.7                      | 速度常数 Speed constant                     | 643          | ±10%      | rpm/V   |
| 1.8                      | 转矩常数 Torque constant                    | 14.85        | Typical   | mNm/A   |
| 1.9                      | 速度/转矩斜率 Speed/torque gradient           | 137          | Typical   | rpm/mNm |

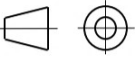
|                        |                                        |                    |         |     |
|------------------------|----------------------------------------|--------------------|---------|-----|
| 2. 齿轮箱参数 Gearhead data |                                        |                    |         |     |
| 2.1                    | 齿轮箱类型 Type of gearhead                 | PLANetary gearhead |         |     |
| 2.2                    | 轴承类型 Bearings type of gearhead         | Ball bearing       |         |     |
| 2.3                    | 名义减速比 Nominal reduction ratio          | 172                | Nominal | X:1 |
| 2.4                    | 精确减速比 Absolute reduction ratio         | 326700/1900        |         |     |
| 2.5                    | 级数 Number of stages                    | 3                  |         |     |
| 2.6                    | 平均空载回差 Average backlash no load        | 1.5                | Typical | °   |
| 2.7                    | 旋转方向 Rotation direction(input/output)  | =                  |         |     |
| 2.8                    | 轴向间隙 Axial play                        | 0~0.1              |         | mm  |
| 2.9                    | 径向间隙 Radial play[mm from flange]       | ≤0.1[5]            |         | mm  |
| 2.10                   | 最大轴向负载 Max.axial load(dynamic)         | 20                 |         | N   |
| 2.11                   | 最大径向负载 Max.radial load[mm from flange] | 50[5]              |         | N   |
| 2.12                   | 最大允许安装力 Max.force for press fits       | 40                 |         | N   |

| 规格 Specifications              |                            | Values  | Tolerance | Unit |
|--------------------------------|----------------------------|---------|-----------|------|
| 3. 整机常规特性 Overall General Data |                            |         |           |      |
| 3.1                            | 额定电压 Nominal voltage       | 24      |           | V dc |
| 3.2                            | 空载转速 No load speed         | 93      | ±10%      | rpm  |
| 3.3                            | 空载电流 No load current       | 150     | ±50%      | mA   |
| 3.4                            | 额定转矩 Rated torque          | 0.52    | Typical   | Nm   |
| 3.5                            | 额定电流 Rated current         | 0.31    | ±10%      | A    |
| 3.6                            | 额定转速 Rated speed           | 85      | ±10%      | rpm  |
| 3.7                            | 最大效率 Maximun efficiency    | 65.6    | Typical   | %    |
| 3.8                            | 堵转转矩 Stall torque          | 15.41   | Typical   | Nm   |
| 3.9                            | 堵转电流 Stall current         | 7.6     | Typical   | A    |
| 3.10                           | 重量 Weight                  | 95      | ±5%       | g    |
| 3.11                           | 使用温度范围 Ambient temperature | -40~+85 |           | °C   |

|              |  |  |  |  |
|--------------|--|--|--|--|
| 4. 其他 Others |  |  |  |  |
| 4.1          |  |  |  |  |
| 4.2          |  |  |  |  |
| 4.3          |  |  |  |  |
| 4.4          |  |  |  |  |

备注 Notes:  
· 所有参数均在20°C给定。大部分参数随温度变化。  
All parameters are given at 20°C. Most of parameters will vary with temperature.  
· 当电机固定在导热支架上时, 热阻通常可降低50%。  
The thermal resistance can be typically decreased of 50% when motor is fixed on a thermally conductive support.  
· 电机电气安全标准符合<GB/T 21418中6.6介电性能>的要求。  
The electrical safety standard of the motor shall meet the requirements of <GB/T21418 6.6 Dielectric properties>.

Unit:mm  
第一角画法  
First angle method  
未注线性与角度尺寸公差  
Tolerances for linear and angular dimensions  
without individual tolerance indications  
GB/T 1804-m eqv ISO 2768-1:m  
未注形状与位置尺寸公差  
Geometrical tolerance for features without  
individual tolerance indications  
GB/T 1184-K eqv ISO 2768-2:K



批准 Approve

标准化 Standardize

工艺 Process Review

审核 Check

设计 Design

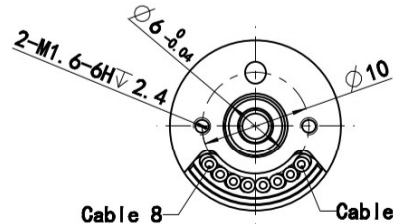
MOONS'  
moving in better ways

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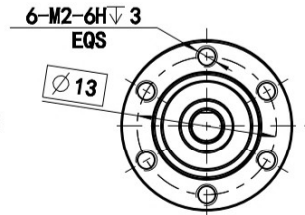
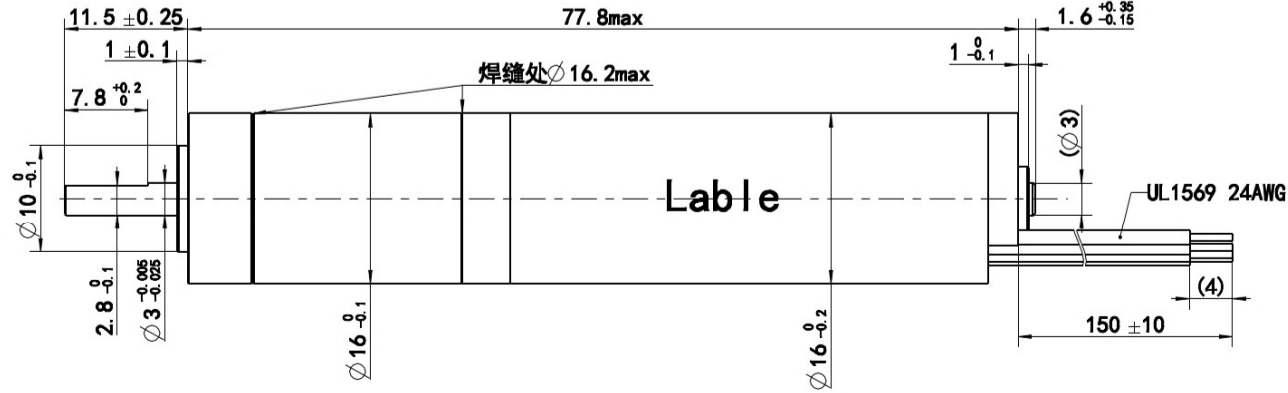
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|------|------------------------|-----------|----------------------------|-------|-----------|------|---------------------------|---------|------------|--------------|
| A0   | 原始版本 Original revision | 2022.4.18 | UNLESS OTHERWISE SPECIFIED | NAMES | SIGNATURE | DATE | Drawing No: 4611180000466 | Rev. A0 | SCALE: 1:1 | Sheet 1 of 2 |
| REV. | 版本记录 REVISION RECORD   | DATE      |                            |       |           |      |                           |         |            |              |

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线束相对于前盖安装孔角度偏差:  $\leq \pm 30^\circ$   
Alignment of cable relative to front mounting holes:  $\leq \pm 30^\circ$



#### Connection

| Cable No. | 信号/Signal                  | 颜色/Color |
|-----------|----------------------------|----------|
| 1         | U                          | 棕/brown  |
| 2         | V                          | 红/red    |
| 3         | W                          | 橙/orange |
| 4         | V <sub>hall</sub> 3~24 VDC | 黄/yellow |
| 5         | GND                        | 绿/green  |
| 6         | Hall 1                     | 蓝/blue   |
| 7         | Hall 2                     | 紫/voilet |
| 8         | Hall 3                     | 灰/grey   |

#### 铭牌 Label Detail

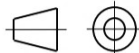
|               |                  |  |
|---------------|------------------|--|
| Motor Type    | ECU16052H24-S002 |  |
| Gearbox No.   | PG16M-0172       |  |
| Work order    | XXXXXXXXX YYYY   |  |
| Serial Number |                  |  |

#### 备注/Note:

- 铭牌用激光标刻在电机机壳表面, 目视水平居中;  
The label is attached to the center of the motor housing.
- 二维码为含电机型号、生产工单号和序列号。  
The QR code includes motor model No., work order No. and serial number.

Unit:mm

第一角画法  
First angle method



未注线性与角度尺寸公差  
Tolerances for linear and angular dimensions  
without individual tolerance indications  
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REV. 版本记录 REVISION RECORD

DATE

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NAMES

SIGNATURE

DATE

Drawing No: 4611180000466

Rev. A0

SCALE: 2:1

Sheet 2 of 2